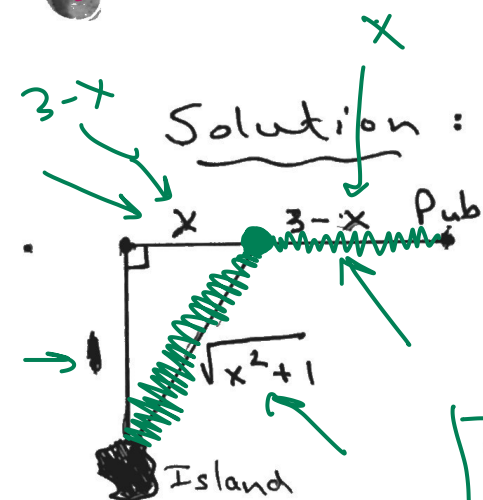
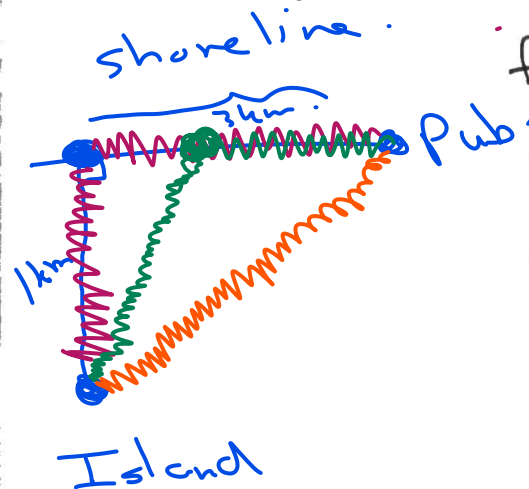


Best Time / Shortest Distance Problems

(1)

Example 1: A man lives on an island 1 km from the shoreline. His favourite pub is 3 km along the shoreline from the point on the shore closest to the island. The man can paddle his canoe at 3 km/hr and can jog 5 km/hr. Determine where he should land in order to reach the pub in the shortest possible time.



Let x be the distance from the point on the shoreline closest to the island.

$$S = \frac{d}{t}$$

$$t = \frac{d}{S}$$

$$\text{Time} = \frac{\text{Distance}}{\text{Speed}}$$

$$\text{Total Time} = T_{\text{paddle}} + T_{\text{jog}}$$

$$\text{Total Time} = \frac{\sqrt{x^2 + 1}}{3} + \frac{3 - x}{5}$$

$$\frac{1}{5}(3 - x)$$

2

$$\left(\frac{\text{Total}}{\text{Time}} \right)' = \frac{1}{3} \cdot \frac{1}{2} \cdot \left(\underline{x^2 + 1} \right)^{-\frac{1}{2}} \cdot \left(\underline{2x} \right) - \frac{1}{5}$$

$$\frac{2x}{6\sqrt{x^2+1}} - \frac{1}{5} = 0$$

$$\frac{x}{3\sqrt{x^2+1}} = \frac{1}{5}$$

$$\frac{x}{\sqrt{x^2 + 1}} = \frac{3}{5}$$

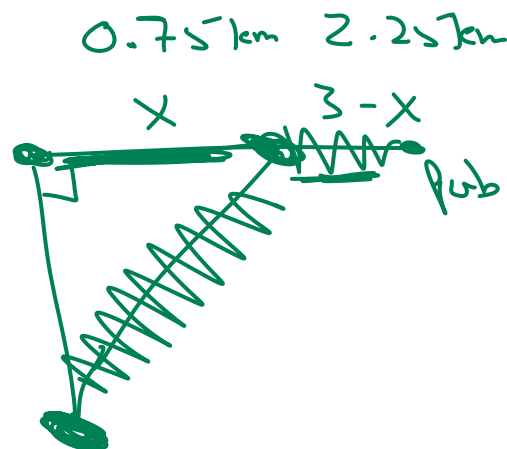
$$\frac{x^2}{x^2 + 1} = \frac{9}{25}$$

$$25x^2 = 9x^2 + 9$$

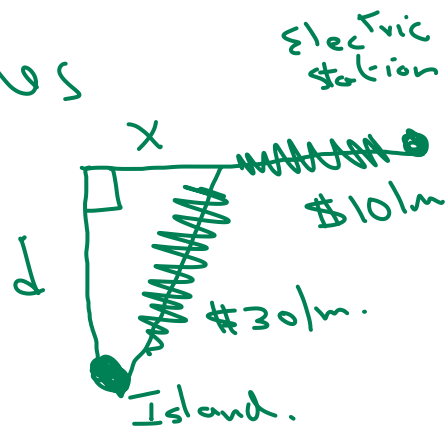
$$16x^2 = 9$$

$$x^2 = \frac{9}{16}$$

$x = \frac{+3}{-4}$ (reject $-\frac{3}{4}$)



1 Square both sides



$\therefore$ He should paddle to a point $\frac{3}{4}$ km from the point closest to the island,

OR

(OR)

∴ He should paddle to a point $\frac{9}{4}$ km from the pub.

$$S = \frac{d}{t} \Rightarrow d = S \cdot t$$

③

Example 2: Mr. Cira leaves Bayview S.S.

N at 1:00 pm and drives due south

At the same time, Mr. Navarro is

heading due west at 90 km/hr

and arrives at Bayview S.S. at

2:00 pm. Determine when they were closest together.

Solution:

Distance = Speed x Time.

BSS $90-90t$ Mr. Navarro

* Mr. Cira's Distance from Bayview S.S. = $80t$

* Mr. Navarro's Distance from Bayview S.S. = $90(1-t)$

= $90-90t$

$$d = \sqrt{(80t)^2 + [90(1-t)]^2}$$

$$d' = \frac{12,800t - 16,200 + 16,200t}{2\sqrt{(80t)^2 + [90(1-t)]^2}} = 0$$

$$d' = 0$$

$$29,000t - 16,200 = 0$$

$$t = 0.559 \text{ hrs } (\text{OR}) \approx 1:34 \text{ pm.}$$

$$0.559 \times 60 = 33.54$$

1:00 pm

2:00 pm

1:34 pm